Population structure in GWAS and SNP heritability: a simulation study on 1000 Genomes chromosome 22

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1 Summary

A genome-wide association study (GWAS) can report false associations when the sample contains ancestry groups that differ in both allele frequency and phenotype (Price et al., 2006). I measured how large this problem is, and how well principal-component (PC) adjustment fixes it, by simulating phenotypes with known causal variants on real genotypes: 2,504 individuals from 26 populations (1000 Genomes Phase 3 (The 1000 Genomes Project Consortium et al., 2015)), chromosome 22 only. In each of two scenarios I simulated 10 replicates with 30 causal variants and a genetic variance share of about 0.4. Scenario A adds noise only; scenario B also adds a mean phenotype shift between continental ancestry groups.

An unadjusted scan is severely inflated in both scenarios (median genomic inflation factor $\lambda = 9.8$ in A and 40.5 in B) and flags on average 17 (A) and 24 (B) distinct megabase windows far from any causal variant. Adjusting for 10 PCs brings λ to about 1.16 and 1.18, and false windows to 0 and 0.1 on average. The adjusted scan recovers about 10 to 12 of 30 causal variants; some of that loss is the correction removing real signal and some is small effects, and I did not separate the two. SNP heritability estimated by REML on a genomic relationship matrix is close to the truth in scenario A (mean bias -0.03 with PCs) and biased upward in scenario B (+0.07 without PCs, +0.04 with PCs). Because λ cannot say which part of the inflation is confounding, I also ran LD score regression against the known truth: its intercept is 1.054 in scenario A, where no confounder exists despite a median λ of 9.8, and 1.805 in scenario B, where one does – but it misclassifies individual replicates often enough that I would not rely on it here (Section 5.5).

This is a simulation on one chromosome with a simplified genetic architecture, so the numbers describe this design, not human GWAS in general.

2 Question

When ancestry is a confounder, (1) how inflated is an unadjusted GWAS, (2) how much of that does PC adjustment remove and at what cost in power, and (3) how biased is a GRM-based SNP-heritability estimate? Simulation lets every one of these be scored against a known truth.

3 Data and design

Genotypes. 1000 Genomes Phase 3, chromosome 22, 2,504 individuals from 5 continental groups (AFR, AMR, EAS, EUR, SAS) (The 1000 Genomes Project Consortium et al., 2015). After QC (biallelic SNPs, MAF $\geq$ 1%, $\leq$ 2% missing) 170,949 variants remain; 97,216 with MAF $\geq$ 5% are used for association scans. I applied no Hardy-Weinberg filter because the sample pools 26 populations, so allele-frequency differences between them create apparent departures from equilibrium in true variants (the Wahlund effect).

Phenotype simulation. For each replicate, 30 causal variants are drawn from common variants with at least 500 kb between them. Standardised effects are drawn from a standard normal distribution and the genetic value g is scaled to variance 0.4. The same causal variants and effects are used in both scenarios for a given replicate.

- *Scenario A:* $y = g + e$, with $e \sim \mathrm{N}(0, 0.6)$.

- *Scenario B:* $y = g + s + e$. s is a fixed shift by continental group (AFR −1, AMR 0, EAS +1, EUR +0.5, SAS −0.5, rescaled to variance 0.2) and $e \sim \mathrm{N}(0, 0.4)$. It stands in for an environmental difference that happens to track ancestry.

The realised heritability ($\mathrm{var}(g)$ / $\mathrm{var}(y)$) is computed per replicate and averages 0.4 in A and 0.42 in B.

4 Methods

Pipeline. PLINK 2 (Chang et al., 2015) for QC, LD pruning (window 200, step 50, $r^2 < 0.2$; 5,855 variants), PCA (10 PCs), the GRM, and the association scans (`--glm` linear regression). R for simulation, scoring and REML. Each replicate is one Slurm array task (`slurm/02_task_array.slurm`) that calls the same script a local loop calls. See Section 7 for what was and was not run.

Association scans. Each replicate is scanned twice: with no covariates, and with 10 PCs as covariates. Genome-wide significance is $p < 5 \times 10^{-8}$.

Scoring against the truth. The genomic inflation factor λ is the median association χ^2 divided by its expected value under the null (0.455) (Bulik-Sullivan et al., 2015). A causal variant counts as *detected* if any significant variant lies within 500 kb. A *false window* is a 1 Mb bin that contains a significant variant farther than 500 kb from every causal variant.

SNP heritability. REML on the GRM from all QC'd variants, using its eigendecomposition so each likelihood evaluation is cheap; implemented in `R/reml.R` and checked against simulated data (`tests/test_reml.R`). GCTA (Yang et al., 2011) has no Apple-silicon build, so I implemented the single-component model directly rather than calling it. The model is fitted without covariates and with the 10 PCs as fixed effects. The truth for comparison is the realised $\mathrm{var}(g)/\mathrm{var}(y)$.

5 Results

5.1 Population structure in the sample

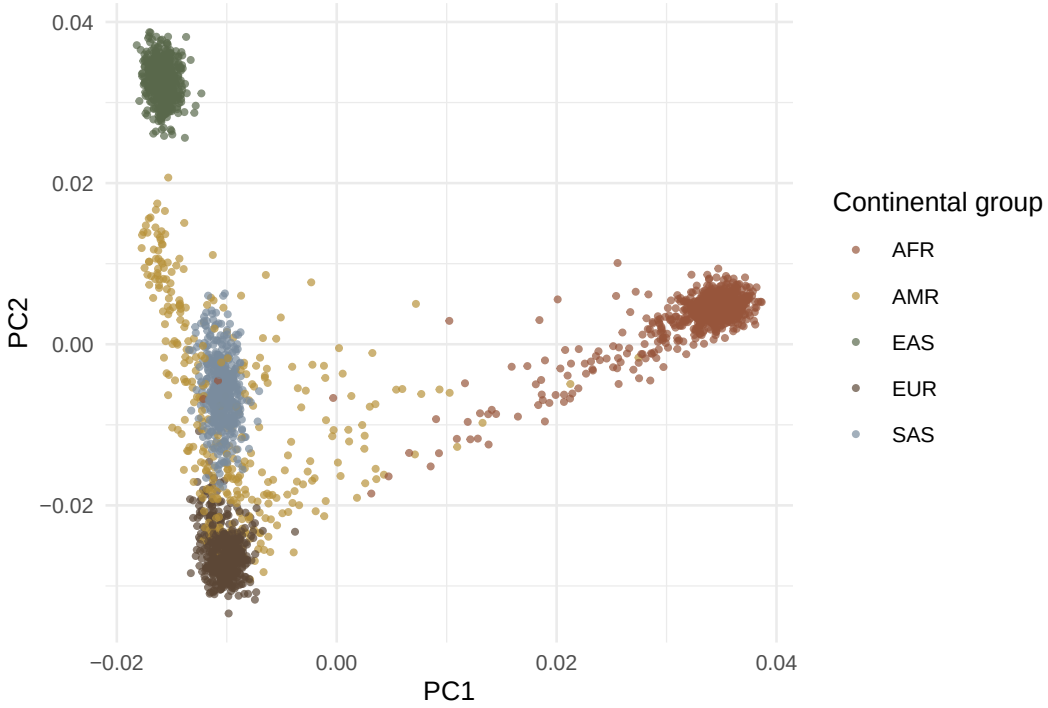


Figure 1: The first two principal components of the LD-pruned genotypes separate the continental groups.

The continental groups separate cleanly on the first two PCs (Figure 1). This is the structure the PCs carry into the adjusted scans. Ten PCs alone explain on average 6.7% of phenotypic variance in scenario A, where no ancestry-linked environmental effect was simulated, and 22.4% in scenario B. In scenario A that 6.7% is almost entirely genetic (ten PCs would explain only about 0.4% of pure noise, 10/2,504): the causal variants have different allele frequencies across groups, so ancestry predicts the genetic value.

5.2 Unadjusted scans are badly inflated, and population shift makes it worse

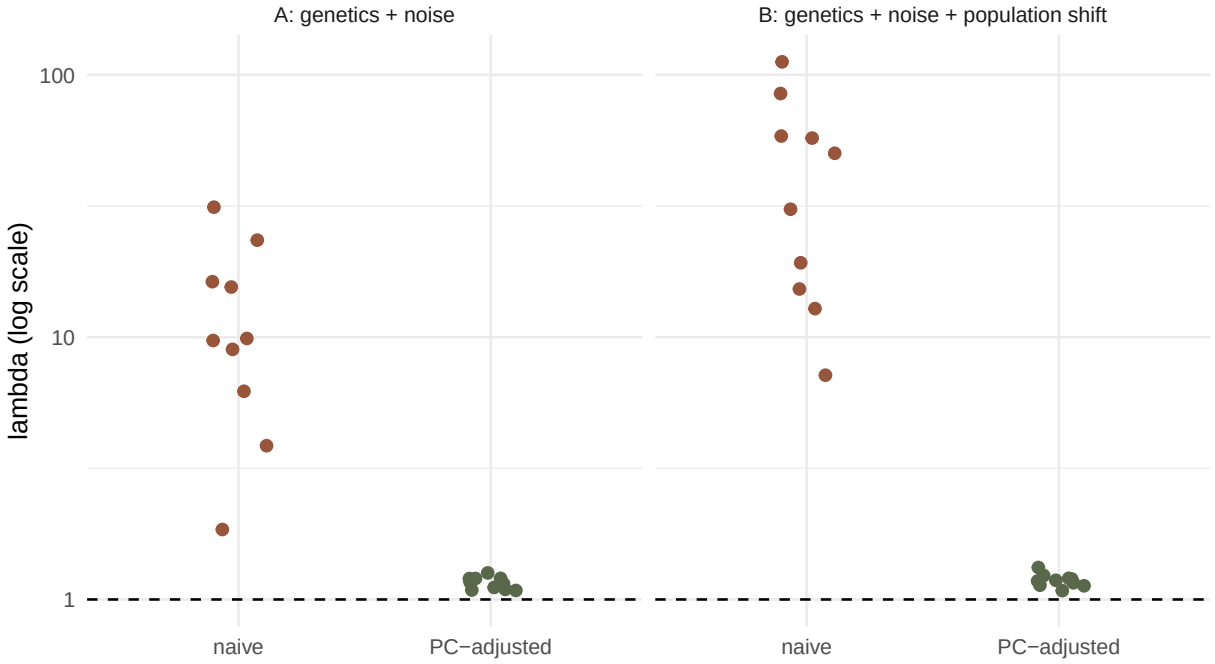


Figure 2: Genomic inflation factor per replicate (log scale). Dashed line: no inflation (lambda = 1).

Table 1: Mean or median (inter-quartile range) over 10 replicates per scenario.

	A: genetics + noise	B: genetics + noise + population shift
Realised h^2	0.40	0.42
lambda, naive	9.8 (6.9-16.1)	40.5 (16.3-58.3)
lambda, PC-adj.	1.16 (1.10-1.20)	1.18 (1.14-1.20)
False windows, naive	17.0	24.0
False windows, PC-adj.	0.0	0.1
Causal detected of 30, PC-adj.	10.2	12.1

Unadjusted λ ranges from 1.8 to 31.3 in A and from 7.2 to 112.2 in B (Figure 2, Table 1). Even scenario A, with no environmental confounding, is inflated: because causal allele frequencies differ between ancestry groups, any variant that is ancestry-informative correlates with the phenotype whether or not it is causal. The replicate-to-replicate spread is large because it depends on which 30 variants happen to be causal. With PCs in the model, λ falls to 1.08-1.26 in A and 1.08-1.33 in B. It does not reach 1, but 30 strong causal variants and a polygenic background also inflate λ legitimately, and λ alone cannot separate that from residual confounding (Bulik-Sullivan et al., 2015). Section 5.5 tests whether LD score regression can.

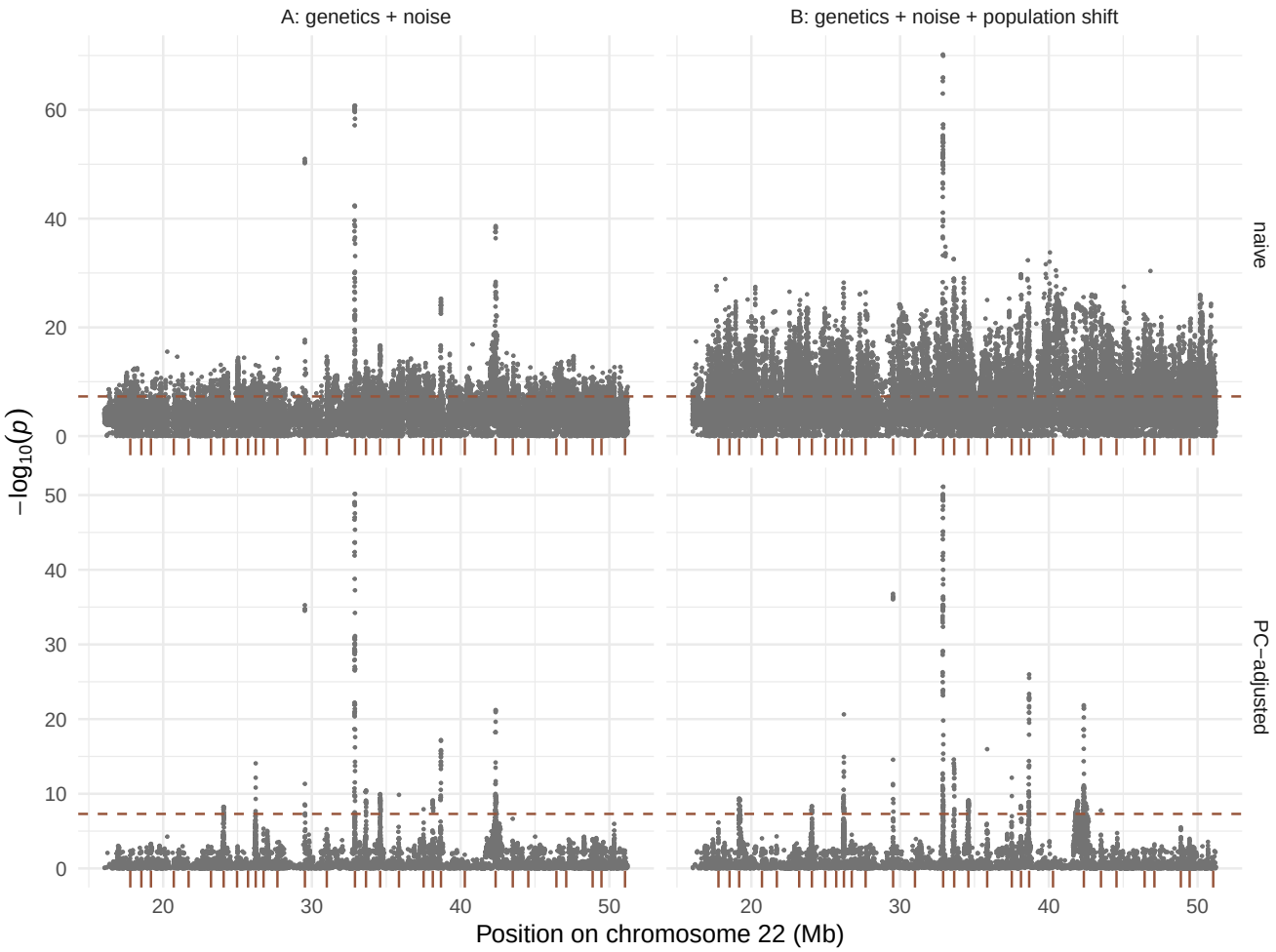


Figure 3: Manhattan plots for replicate 1 of each scenario (thinned: all p < 0.01 plus 5% of the rest). Dashed line: p = 5e-8. Red ticks mark true causal variants.

5.3 The price of adjustment

The unadjusted scan "detects" nearly all 30 causal variants (26.9 on average in A and 30 in B), but this number cannot be read as power: with thousands of significant variants spread along the chromosome (Figure 3), a window around any causal variant will almost always contain one by chance. Detection is only meaningful for the adjusted scan, where it is about 10.2 of 30 in A and 12.1 in B. Some of that loss is the correction removing real signal that is correlated with ancestry, and some is simply that these effects are small for a sample of 2,504. I did not separate the two, so the adjusted detection rate should not be read as the power cost of PC adjustment alone.

5.4 SNP heritability

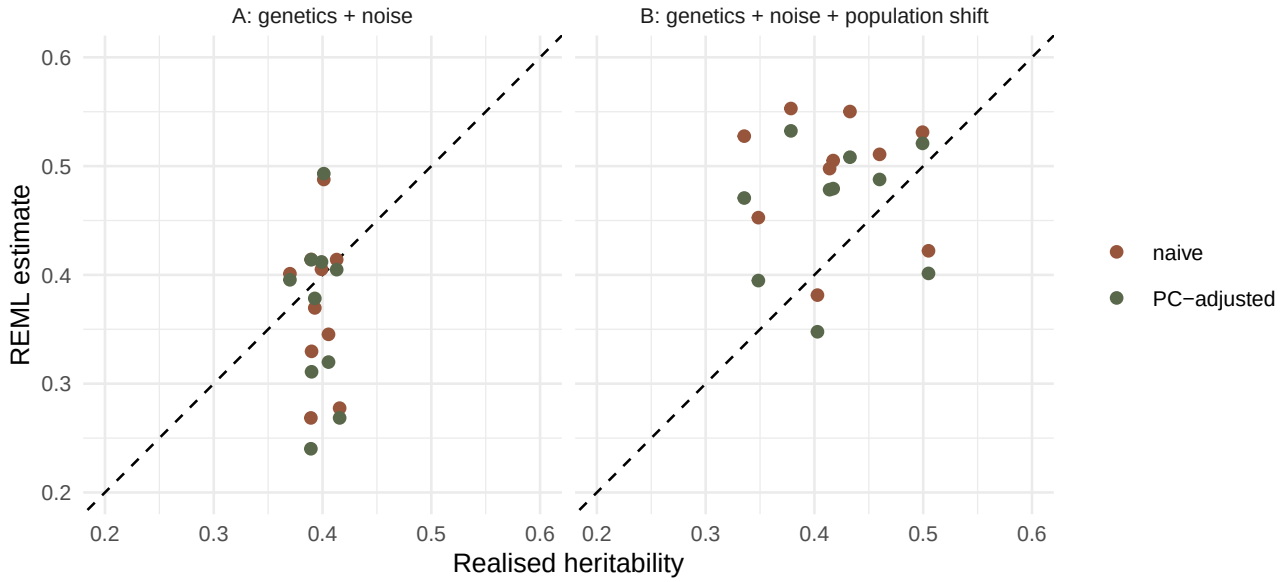


Figure 4: REML SNP-heritability estimates against the realised truth, one point per replicate. Dashed line: estimate = truth.

In scenario A the estimates scatter around the truth with a small mean bias (-0.025 without PCs, -0.033 with) and a standard deviation of about 0.07. In scenario B the unadjusted estimate is biased upward by +0.074 on average: the ancestry-linked shift is captured as "genetic" because the GRM encodes ancestry. Adding PCs reduces the bias to +0.043 but does not remove it (Figure 4). With 10 replicates per scenario, these mean biases carry sizeable uncertainty, since the per-replicate standard deviation of the bias is about 0.08.

5.5 Can LD score regression tell confounding from polygenicity?

λ cannot separate the two, which is why the section above refuses to read the adjusted λ as a confounding estimate. LD score regression (Bulik-Sullivan et al., 2015) claims it can: regressing each variant's χ^2 on its LD score ℓ_j,

$$\mathrm{E}[\chi_j^2] = 1 + Na + \frac{Nh_g^2}{M}\,\ell_j,$$

the slope carries polygenic signal and the intercept $1 + Na$ carries confounding. A simulation can test that claim, because here the truth is known: scenario A has no environmental confounder at all, scenario B has one.

I computed LD scores for all 97,216 common variants from these same 2,504 samples (SNPs within 1 Mb either side, unbiased r^2; mean $\ell = 158.49$, median 126.4), then ran the regression with LDSC's own weights, its $\chi^2 \le 80$ outlier rule and a 200-block genomic jackknife for the standard error (`scripts/04_ldsc.sh`, `R/ld_scores.R`, `R/ldsc.R`).

Table 2: LD score regression over 10 replicates per scenario. "Calls confounding" counts replicates whose intercept exceeds 1 by more than two jackknife standard errors. In scenario A that is a false positive by construction.

Scenario	Model	Median lambda	Median intercept	Intercept IQR	Mean SE	Calls confounding
A: no confounder	unadjusted	9.80	1.054	0.81-1.36	0.103	3/10
A: no confounder	10 PCs	1.16	1.135	1.10-1.18	0.061	6/10

Scenario	Model	Median lambda	Median intercept	Intercept IQR	Mean SE	Calls confounding
B: confounder	unadjusted	40.51	1.805	1.22-2.30	0.176	7/10
B: confounder	10 PCs	1.18	1.146	1.12-1.20	0.070	7/10

The intercept carries information λ does not. Scenario A has no confounder, yet its unadjusted λ has a median of 9.8; read on its own that would say the scan is badly confounded. The intercept instead reads 1.054. In scenario B, where a confounder was simulated, it rises to 1.805. The intercept is also not simply tracking λ: within scenario A the two are *negatively* rank-correlated (-0.66), and the three most inflated A replicates have some of the lowest intercepts.

It is not reliable enough to use per study on this design. In scenario A the intercept ranges 0.573 to 1.989 and 3 of 10 replicates call confounding that does not exist; in scenario B 3 replicates miss a confounder that does. Values below 1 are outside the model, and several appear.

After PC adjustment it is biased upward. Both scenarios give intercepts near 1.135 and 1.146, significantly above 1 in 6 and 7 of 10 replicates – including scenario A, where after adjustment there is nothing left to find. On this design the intercept should not be read as evidence of residual confounding.

The slope is unusable here, as expected. Implied h^2 has a median of 1.716 (A) and 5.324 (B) against a truth near 0.4. Thirty large-effect variants on one chromosome are the opposite of the polygenic architecture LDSC assumes, and the LD scores come from the same 26-population sample rather than an ancestry-matched reference panel, which inflates r^2 between ancestry-informative variants. Only the intercept is interpretable, and only loosely.

6 Limitations

- **One chromosome.** All 30 causal variants and the GRM are on chromosome 22. Real traits spread causal variants across the genome and have many more of them, with far smaller effects, so this design has higher per-variant power and a more extreme ancestry-genetics correlation than a genome-wide polygenic trait.
- **Simplified architecture.** Effects are independent of allele frequency and there is no LD-dependent heritability, which real traits are known to have. Heritability estimates may be sensitive to that choice; I did not vary it.
- **Few replicates.** 10 per scenario. Medians and IQRs are shown for λ because its distribution is heavily skewed; mean biases in heritability are uncertain to roughly the standard deviations above divided by $\sqrt{10}$.
- **The "detected" definition** (a significant variant within 500 kb) is generous and undefined for an inflated scan; it should be treated as descriptive.
- **Adjustment choice.** Ten PCs were used in every case; I did not vary the number or compare with a mixed-model association approach.
- **LD score regression** is run on chromosome 22 only, with LD scores from the sample being analysed rather than a matched reference panel, and on a 30-variant architecture. Its intercept separates the two scenarios at the median but misclassifies individual replicates in both directions, so Section 5.5 reports it as a check on the method, not as a tool I would rely on.
- **Simulated phenotypes only.** A natural next step is to apply the same scoring to public GWAS summary statistics, where the truth is unknown and only calibration can be checked.

7 Reproducibility and portability

The full pipeline is in the repository: `scripts/` (bash), `R/`, `slurm/` and `run_local.sh`.

Item	Status
QC, 20 simulation replicates, scoring and this report	**Run end to end** on macOS (Apple silicon), under the system bash 3.2 (`/bin/bash`), with PLINK 2.0 and R 4.5
`scripts/env.sh` under bash and zsh 5.9	**Run:** `tests/test_shells.sh` checks both shells resolve identical paths and task mapping
LD score regression (`scripts/04_ldsc.sh`)	**Run end to end** on macOS. LD scores for all 97,216 variants take about 45 minutes; the 40 scans it re-runs take seconds. Syntax-checked under bash and zsh
Slurm job scripts (`slurm/*.slurm`, dependency chain in `submit_all.sh`)	**Syntax-checked (`bash -n`) only.** They call the same `scripts/run_task.sh` that the local loop ran, but **have not been submitted to a Slurm scheduler.** The `#SBATCH` resource requests are my estimates, not measured
Linux and Windows	**Smoke tests only.** A GitHub Actions workflow ran the bash/zsh environment test, a syntax check of every shell script and the REML self-test on Ubuntu, macOS and Windows, and all three passed (28 Oct 2026). The full pipeline (PLINK 2 QC, simulation, scans) was **not** run on Linux or Windows, and `run_local.ps1` has not been run.

The scripts use no bash 4 features, because macOS ships bash 3.2.

8 Provenance

Genotypes are public (1000 Genomes) and phenotypes are simulated. The code was written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding the results. My STAT7306 coursework used a Slurm cluster, GCTA and PLINK on assessed data; none of that data, code or text is used here, and the design, data and simulation are different.

References

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